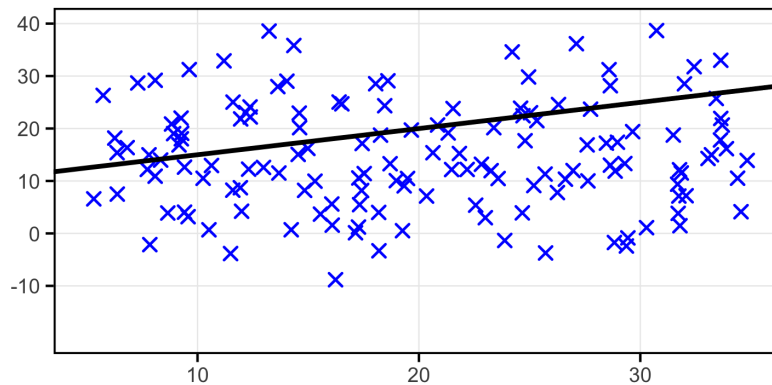


Regression and Analysis of Variance

Fall 2025, Math 533 Course Notes

McGill University

**Everything is linear
if you are brave enough**



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Despite all efforts, there may still be some typos, unclear explanations, etc. If you find potential mistakes, or any suggestions regarding concepts or formats, etc., feel free to reach out to the author at zhangjohnson729@gmail.com.

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Review of Asymptotic Statistics

1.1 Random Variables and Convergence

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the probability space, where Ω is some arbitrary non-empty set (we usually denote as the sample space). $\mathcal{F}$ is another set which contains a collection of subsets of Ω that satisfies: (i) $\Omega \in \mathcal{F}$; (ii) Closed under set compliments; (iii) Closed under countable unions. $\mathcal{F}$ is also called a σ -algebra of Ω . We denote $\mathfrak{B}(\mathbb{R})$ as the σ -algebra generated by all the open sets of $\mathbb{R}$, which is called Borel σ -algebra. $\mathbb{P}$ is the probability measure (a set function) $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ such that: (i) $\mathbb{P}(\Omega) = 1$; (ii) If $\{X_i\}_{i=1}^{\infty} \subseteq \mathcal{F}$ and $X_i \cap X_j = \emptyset$ whenever $i \neq j$ then $\mathbb{P}\left(\bigcup_{i=1}^{\infty} X_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(X_i)$.

A random variable X is also a function $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathfrak{B}(\mathbb{R}))$ such that $\forall B \subseteq \mathfrak{B}(\mathbb{R})$, its pre-image $X^{-1}(B) \subseteq \mathcal{F}$. We will work with a sequence of random variables $\{X_i\}_{i=1}^{\infty}$ defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. There are four types of convergences we are interested in, namely **weak convergence**, **convergence in probability**, **convergence in L^p** , **convergence almost surely**. We write $X_n \xrightarrow{L} X$, meaning X_n converges to X weakly, (or in law, in distribution) if $\mathbb{P}(X_n \leq x) \rightarrow \mathbb{P}(X \leq x)$ for all x such that $x \mapsto \mathbb{P}(X \leq x)$ is continuous, or by saying $F_n(x) \rightarrow F(x)$ where F represents the cumulative distribution function. We write $X_n \xrightarrow{P} X$, meaning X_n converges to X in probability, if $\forall \varepsilon > 0, \mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$. We write $X_n \xrightarrow{L^p} X$, meaning X_n converges to X in L^p ($p \geq 1$), if $\mathbb{E}|X_n - X|^p \rightarrow 0$. Lastly if X_n converges to X almost surely, we have $\mathbb{P}(\lim_{n \rightarrow \infty} X_n = X) = 1$ and denote as $X_n \xrightarrow{a.s.} X$.

Theorem

Theorem 1. (Markov's Inequality) Let $X \geq 0$ a.s, then for all $t \geq 0$,

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t} \quad (1.1)$$

Proof.

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E}[X \cdot \mathbf{1}\{X \geq t\}] + \mathbb{E}[X \cdot \mathbf{1}\{X < t\}] \\ &\geq \mathbb{E}[X \cdot \mathbf{1}\{X \geq t\}] \\ &= t\mathbb{P}(X \geq t). \end{aligned}$$

■

We may use Markov's inequality to deduce Chebyshev's inequality: If $\mathbb{E}[X^2] < \infty$ then $\forall t > 0$,

$$\mathbb{P}(|X - \mathbb{E}X| > t) \leq \frac{\mathbb{V}(X)}{t^2} \quad (1.2)$$

where we have

$$\mathbb{P}(|X - \mathbb{E}X| > t) = \mathbb{P}(|X - \mathbb{E}X|^2 > t^2) \leq \frac{\mathbb{V}(X)}{t^2} \quad (1.3)$$

where by definition $\mathbb{E}[|X - \mathbb{E}X|^2] := \mathbb{V}(X)$.

In general, the different modes of convergence can be related by the following diagram:

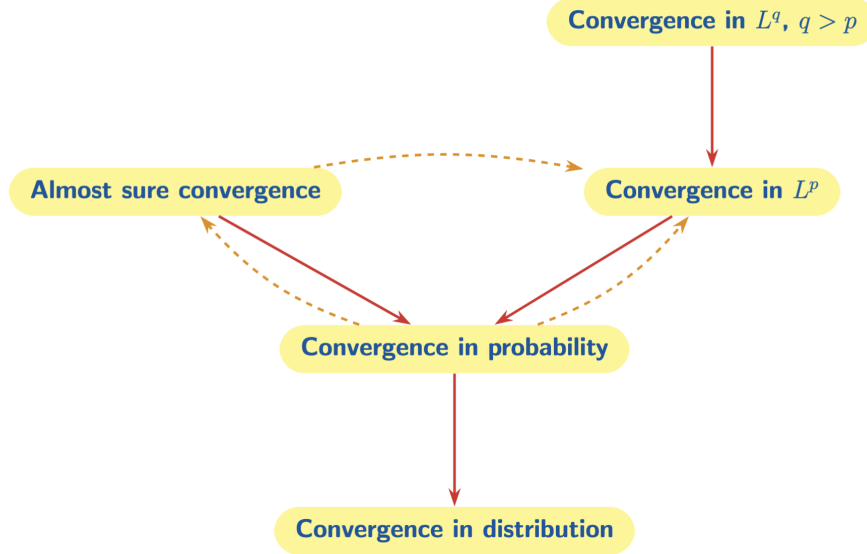


Figure 1: The diagram shows the relations between different modes of convergence. The arrows in red means direct implication without any further condition, while arrows in orange will hold if extra conditions are given. The general structure is that, convergence in probability implies convergence almost surely along a subsequence; Convergence in probability with uniform integrability would imply convergence in L^p ; Convergence almost surely when dominated convergence theorem applies will imply convergence in L^p .

The next few theorems will show the proofs for some arrows. Denote $\{X_i\}_{i=1}^\infty$ be a sequence of random variables defined on $(\Omega, \mathcal{F}, \mathbb{P})$.

Theorem

Theorem 2. If $X_n \xrightarrow{P} X$, then there exists a subsequence n_k of $\mathbb{N}$ such that $X_{n_k} \xrightarrow{a.s.} X$.

Proof. Assume $X_n \xrightarrow{P} X$, then $\forall \varepsilon > 0$, $\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$ as $n \rightarrow \infty$, meaning that $\forall k \geq 1$, $\exists n_k$ such that $\mathbb{P}(\{|X_{n_k} - X| > 1/k\}) \leq 1/k^2$, denote $A_k := \{|X_{n_k} - X| > 1/k\}$, then by *Borel-Cantelli Lemma*, we have

$$\mathbb{P}\left(\bigcap_{l=1}^{\infty} \bigcup_{k=l}^{\infty} A_k\right) = \lim_{l \rightarrow \infty} \mathbb{P}\left(\bigcup_{k=l}^{\infty} A_k\right) \leq \lim_{l \rightarrow \infty} \sum_{k=l}^{\infty} \mathbb{P}(A_k) = 0, \quad (1.4)$$

meaning that for almost everywhere, $\exists l$, such that $\forall k \geq l : |X_{n_k} - X| \leq 1/k$, which means that for almost everywhere, $\lim_{k \rightarrow \infty} |X_{n_k} - X| = 0$, thus $X_{n_k} \xrightarrow{a.s.} X$. ■

Theorem

Theorem 3. If $X_n \xrightarrow{L^p} X$, then $X_n \xrightarrow{P} X$.

Proof. The proof is straightforward, we have

$$\mathbb{P}\{|X_n - X| > \varepsilon\} = \mathbb{P}\{|X_n - X|^p > \varepsilon^p\} \leq \frac{\mathbb{E}|X_n - X|^p}{\varepsilon^p} \rightarrow 0. \quad (1.5)$$

■

1.2 Law of Large Numbers and Central Limit Theorem

Assume we have a sequence of random variables $\{X_i\}_{i=1}^{\infty} \stackrel{i.i.d}{\sim} \mathbb{P}_x$, then in this section we will introduce some important theorems in probability.

Theorem

Theorem 4. (Weak Law of Large Numbers) Assume $\mathbb{E}|X| < \infty$, then

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mathbb{E}[X]. \quad (1.6)$$

Proof. Our task is much easier if we assume $\mathbb{E}|X|^2 < \infty$. Then Chebyshev's inequality states that

$$\begin{aligned} \mathbb{P}\left\{\left|\bar{X}_n - \mathbb{E}[X]\right| > \varepsilon\right\} &\leq \frac{\mathbb{V}(\bar{X}_n)}{\varepsilon^2} \\ &= \frac{\mathbb{V}(X)}{n\varepsilon^2} \rightarrow 0. \end{aligned}$$

■

In fact a stronger statement can be shown, known as the Strong Law of Large Numbers (SLLN), where

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s} \mathbb{E}[X]. \quad (1.7)$$

So far don't think about the proof of (1.6). If you really want some torture, check out Probability Theory by Daniel Stroock, Section 1.4.. Next we introduce a technical lemma to prove central limit theorem:

Lemma

Lemma 1. (Levy Continuity Theorem) The characteristic function of X is defined as

$$\mathbb{1}_X(t) := \mathbb{E}[\exp(itX)]. \quad (1.8)$$

Then $X_n \xrightarrow{L} X$ iff $f_{X_n}(t) \xrightarrow{\text{pointwise}} f_X(t)$ for all $t \in \mathbb{R}$.

Now we state the central limit theorem:

Theorem

Theorem 5. Let $\{X_i\}_{i=1}^{\infty} \stackrel{i.i.d}{\sim} f$ and assume $\mathbb{E}[X^2] < \infty$, then

$$\sqrt{n}(\bar{X}_n - \mathbb{E}[X]) \xrightarrow{L} N(0, \mathbb{V}(X)). \quad (1.9)$$

Proof. WLOG assume $\mathbb{E}X = 0, \mathbb{V}(X) = 1$ then

$$\begin{aligned} \mathbb{1}_{\sqrt{n}\bar{X}_n}(t) &= \mathbb{E}\left[\exp\left(\sqrt{n}it\bar{X}_n\right)\right] \\ &= \mathbb{E}\left[\exp\left(\frac{it(X_1 + \dots + X_n)}{\sqrt{n}}\right)\right] \\ &= \left(\mathbb{E}\left[\exp\left(\frac{itX}{\sqrt{n}}\right)\right]\right)^n \\ &= \left[\mathbb{1}_X\left(\frac{t}{\sqrt{n}}\right)\right]^n. \end{aligned}$$

Then a Taylor expansion around 0 will yield

$$\begin{aligned} \mathbb{1}_X\left(\frac{t}{\sqrt{n}}\right) &= \mathbb{1}_X(0) + \mathbb{1}'_X(0) \cdot \frac{t}{\sqrt{n}} + \mathbb{1}''_X \cdot \frac{t^2}{2n} + o\left(\frac{1}{n}\right) \\ &= 1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right) \end{aligned}$$

This is because

$$\mathbb{1}'_X(t) \Big|_{t=0} = \frac{d}{dt} \Big|_{t=0} \int_B f(x) \cdot \exp(itX) dx \quad (1.10)$$

$$= \int_B \frac{d}{dt} \Big|_{t=0} f(x) \cdot \exp(itX) dx \quad (1.11)$$

$$= \int_B ix f(x) \cdot \exp(itX) \Big|_{t=0} dx \quad (1.12)$$

$$:= i\mathbb{E}[X] \quad (1.13)$$

$$= 0. \quad (1.14)$$

A similar statement can be drawn: $\mathbb{1}''_X(0) = i^2\mathbb{E}[X^2] = -1$. Use the fact that $\left(1 + \frac{x}{n}\right)^n \sim e^x$ for

large n , we have

$$\begin{aligned}\mathbb{1}_{\sqrt{n}\bar{X}_n}(t) &= \left[\mathbb{1}_X \left(\frac{t}{\sqrt{n}} \right) \right]^n \\ &= \left[1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right) \right]^n \\ &= \exp \left(-\frac{t^2}{2} \right).\end{aligned}\tag{1.15}$$

By the uniqueness of the characteristic function, (1.15) is the characteristic function of $N(0,1)$. ■

1.3 Convergence as Functions of Random Variables

The main theorems we will introduce are continuous mapping theorem and Slutsky's theorem:

Theorem

Theorem 6. (Continuous Mapping Theorem) Assume $X_n \xrightarrow{m} X$, where m represents any mode of convergence (i.e in distribution, in probability, almost surely, in $\mathcal{L}^p$), and let f be continuous at x , then $f(X_n) \xrightarrow{m} f(X)$.

Proof. I am not proving this. ■

Theorem

Theorem 7. (Slutsky's Theorem) Assume $X_n \xrightarrow{L} X$, $Y_n \xrightarrow{P} c$ for some constant c , then: (i) $X_n + Y_n \xrightarrow{L} X + c$; (ii) $X_n Y_n \xrightarrow{L} cX$; (iii) $X_n/Y_n \xrightarrow{L} X/c$.

Proof. I am not proving this. ■

1.4 Some Random Facts

Proposition

Proposition 1. (Law of Total Expectation) Let X, Y be random variables defined on $(\Omega, \mathcal{F}, \mathbb{P})$ and g be a measurable function, then $\mathbb{E}[g(X)] = \mathbb{E}[\mathbb{E}[g(X)|Y]]$.

Proof. Wlog, let X, Y be continuous with density function $f(x), g(y)$, then

$$\mathbb{E}[g(X)] = \int \mathbb{E}[g(x)|Y = y] dy = \int \int g(x) f_{X|Y}(x|y) dy dx := \mathbb{E}[\mathbb{E}[g(X)|Y]].\tag{1.16}$$

■

Proposition

Proposition 2. (Variance Decomposition) Let X be square integrable, then

$$\mathbb{V}(X) = \mathbb{E}[\mathbb{V}(X|Y)] + \mathbb{V}(\mathbb{E}[X|Y]). \quad (1.17)$$

Proof. By direct computation, one would have

$$\mathbb{E}[\mathbb{V}(X|Y)] + \mathbb{V}(\mathbb{E}[X|Y]) = \mathbb{E}[\mathbb{E}[X^2|Y] - (\mathbb{E}[X|Y])^2] + \mathbb{E}[\mathbb{E}[X|Y]]^2 - (\mathbb{E}[\mathbb{E}[X|Y]])^2 \quad (1.18)$$

$$= \mathbb{E}[X^2] - \mathbb{E}(\mathbb{E}[X|Y])^2 + \mathbb{E}(\mathbb{E}[X|Y])^2 - (\mathbb{E}[X])^2 \quad (1.19)$$

$$:= \mathbb{V}(X). \quad (1.20)$$

where in (1.19) we used the law of total expectation. ■

Linear Regression & Regression Analysis

2.1 An Introduction

Our basic set up: Let $(x, y) \in \mathbb{R}^p \times \mathbb{R}$ be a random vector defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, let $\mathbb{P}_{x,y}$ denote the joint distribution of x, y . Without specification, all random variables are square-integrable, that is, $\mathbb{E}|X|^2 < \infty$.

Definition

Definition 1. The coordinates of x , denoted $\{x_j\}_{j \in [p]}$ is called the **covariates**, or independent variables; y is called the **dependent variable**, or the **response**, or the variable of interest; p is the dimension of the covariates.

In this course, we will consider all response y are continuous. Recall that a numeric variable is said to be discrete if its support is at most countable; otherwise it is said to be continuous. The continuous response we can think of are income, weight. For covariates x , it normally has the following types:

- (i) Quantitative continuous covariates, like income, weight, etc.
- (ii) Transformations of quantitative inputs, like different functions of a original covariate. Say $x_2 = x_1^2, x_3 = \log(x_1)$, etc.
- (iii) Functions of original covariates, say $x_3 = x_1 + x_2$.
- (iv) Categorical covariates, usually coded as dummy variables. Like for gender, we use $X = 1$ for male and $X = 0$ for female, for example.

The goal of regression is to analysis and find the relation between the covariates x and the response y . We recall that $\mathbb{P}_{x,y}$ is the joint distribution of x, y , which can be written as

$$\mathbb{P}_{x,y} = \mathbb{P}_x \cdot \mathbb{P}_{y|x} \tag{2.1}$$

through a conditional probability argument, in above $\mathbb{P}_x$ is the marginal distribution of all covariates and $\mathbb{P}_{y|x}$ is the conditional distribution of y given x . It is not easy to find the conditional distribution, but we can work with conditional expectation $\mathbb{E}[y|x]$ instead.

Theorem

Theorem 8. Let $\mathcal{M}$ denote the set of measurable functions from $\mathbb{R}^p$ to $\mathbb{R}$ and denote by m the function $m : \mathbf{u} \rightarrow \mathbb{E}[y|x = \mathbf{u}] \in \mathcal{M}$, then

$$m = \arg \min_{f \in \mathcal{M}} \mathbb{E}[\{y - f(\mathbf{x})\}^2]. \quad (2.2)$$

This is known as the best prediction property under the L^2 risk.

Proof. We have

$$\begin{aligned} \mathbb{E}[\{y - f(\mathbf{x})\}^2] &= \mathbb{E}[\{y - m(\mathbf{x}) + m(\mathbf{x}) - f(\mathbf{x})\}^2] \\ &= \mathbb{E}[\{y - m(\mathbf{x})\}^2] + \mathbb{E}[\{m(\mathbf{x}) - f(\mathbf{x})\}^2] + 2\mathbb{E}[\{y - m(\mathbf{x})\} \cdot \{m(\mathbf{x}) - f(\mathbf{x})\}], \end{aligned}$$

where

$$\begin{aligned} \mathbb{E}[\{y - m(\mathbf{x})\} \cdot \{m(\mathbf{x}) - f(\mathbf{x})\}] &= \mathbb{E}[\mathbb{E}[\{y - m(\mathbf{x})\} \cdot \{m(\mathbf{x}) - f(\mathbf{x})\} | \mathbf{x}]] \\ &= \mathbb{E}[\{m(\mathbf{x}) - f(\mathbf{x})\} \cdot \mathbb{E}[\{y - m(\mathbf{x})\} | \mathbf{x}]] \\ &= \mathbb{E}[\{m(\mathbf{x}) - f(\mathbf{x})\} \cdot (\mathbb{E}[y | \mathbf{x}] - m(\mathbf{x}))] \end{aligned}$$

and by definition, $m(\mathbf{x}) = \mathbb{E}[y | \mathbf{x}]$ so the above term will become zero. Hence now we have

$$\mathbb{E}[\{y - f(\mathbf{x})\}^2] = \mathbb{E}[\{y - m(\mathbf{x})\}^2] + \mathbb{E}[\{m(\mathbf{x}) - f(\mathbf{x})\}^2] \quad (2.3)$$

as a function of f , so it is easy to see it will attain the minimum when $f(\mathbf{x}) = m(\mathbf{x}) = \mathbb{E}[y | \mathbf{x}]$. ■

We can also prove the theorem using projection theorem:

Theorem

Theorem 9. Let $\mathcal{H}$ be a Hilbert space, $\mathcal{W} \subseteq \mathcal{H}$ closed, and $\forall x \in \mathcal{H}$ there is a unique $P_{\mathcal{W}}(x) \in \mathcal{W}$ such that

$$\|x - P_{\mathcal{W}}(x)\| = \inf_{y \in \mathcal{W}} \|x - y\|. \quad (2.4)$$

Then consider $\mathcal{M}$ denote the set of all measurable functions from $\mathbb{R}^p$ to $\mathbb{R}$, and $L^2(\mathcal{M}) := \{z = f(\mathbf{x}), f : \mathbb{R}^p \rightarrow \mathbb{R}\} \subseteq L^2$, we know L^2 is a Hilbert space with inner product defined by $\{Z_1, Z_2\} = \mathbb{E}[Z_1 Z_2]$ and then using projection theorem $\mathbb{E}[(y - P_{\mathcal{M}}(y))f(\mathbf{x})] = 0$ for all f , and we choose such g and m then $\mathbb{E}[(m(\mathbf{x}) - g(\mathbf{x}))f(\mathbf{x})] = 0$, which is $\mathbb{E}[(m(\mathbf{x}) - g(\mathbf{x}))^2] = 0$ which implies $m(\mathbf{x}) = g(\mathbf{x})$ a.e.

We may also write $y = m(\mathbf{x}) + (y - m(\mathbf{x})) = m(\mathbf{x}) + \varepsilon$ as a derivation, and ε is defined as the noise term, $m(\mathbf{x})$ is the regression function, if $\varepsilon = 0$ almost surely, the model is said to be noiseless.

In practice, we do not have access to the true conditional distribution $\mathbb{P}_{y|x}$, what we have is an observable sample $D_n = \{(x_1, y_1), \dots, (x_n, y_n)\}$ of n i.i.d random variables with joint distribution $\mathbb{P}_{x,y}$. We say D_n is the observed sample, and n is the sample size. (Unless stated, we will assume $p < n$). The goal of regression is to estimate and understand the relationship between x and y , using only the observed sample.

The covariates can choose to be either fixed or random. For fixed design the covariates are constant and they do not change; while in a random design the covariates are random. Both methods are valid in different contexts: Fixed design regression is especially appropriate when the data has been generated rather than observed; while random design regression allows for a more general treatment and is especially suitable for non-experimental sciences such as econometrics. A example is that, suppose we want to study the relationship between the person's age (covariate) and salary (the response y), then a fixed covariate design would be choose people from different ages by design; however a random design would be choose people at random and then we have access to their age. This selection process is treated as random.

2.2 Least Square Loss

Recall that if we want to use $f(x)$ to model the response y , we have the mean squared error (MSE) defined by

$$\mathbb{E}[(y - f(x))^2] = \mathbb{E}[(y - m(x))^2] + \mathbb{E}[(m(x) - f(x))^2] \quad (2.5)$$

where $m(x) = \mathbb{E}[y|X = x]$, then based on our observable data $\mathcal{D}_n$, how can we find such a function f ? Note that the first term on the right hand side above does not depend on f and it suffices to find

$$f^* = \arg \min_{f \in \mathcal{M}} \mathbb{E}[(y - f(x))^2] \quad (2.6)$$

Also note that in reality we do not have access to the moment either, since we don't know the true distribution. But instead we can replace by its empirical estimate:

$$\hat{f} := \arg \min_{f \in \mathcal{M}} \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 \quad (2.7)$$

Definition

Definition 2. Let $f \in \mathcal{M}$ be a given function, the least square loss of f over $\mathcal{D}_n$ is

$$\mathcal{L}_n(f) := \frac{1}{n} \sum_{i=1}^n \{y_i - f(x_i)\}^2. \quad (2.8)$$

Now there is another issue: If $\mathbb{P}_x$ is absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}^p$, then $\mathbb{P}(X_i \neq X_j) = 1$ almost surely for $i \neq j$. Then, each x_i must be distinct, and then we can in fact find a class of polynomials $\mathcal{P}$ such that each $p(x) \in \mathcal{P}$ will passes through all x_i exactly, then we can minimize the term to 0 exactly. Then we might have some “bumpy” polynomials that jumps back and forth and hence it is not a good represent of our data. Remember our goal is to find such a f such that, if we have a new input x_{n+1} , what would be a good fit for its y_{n+1} , based on

the model we designed. So by using those “bumpy” polynomials is absolutely not a good choice. This phenomenon is known as **overfitting**. To avoid the problem of interpolating functions, we can then define a subclass $\mathcal{C} \subseteq \mathcal{M}$ and indeed try to find a minimizing f in the class $\mathcal{C}$. This class can depend on n . As we can see, if $\mathcal{C}_n$ gets closer and closer to $\mathcal{M}$, we will get more estimation errors because of the problem of overfitting I discussed; also if $\mathcal{C}_n$ is too small, then although we might be able to find f under this class easily, but it might be a bad approximation and not we want. The following theorem tells us we can indeed bound the MSE by the “estimation error” and the “approximation error”:

Theorem

Theorem 10. Let $\mathcal{C}_n$ be a class of functions depending on $\mathcal{D}_n$, if $\hat{m}_n := \hat{m}(\cdot, \mathcal{D}_n)$ satisfies

$$\hat{m}(\cdot, \mathcal{D}_n) := \arg \min_{f \in \mathcal{C}_n} \frac{1}{n} \sum_{i=1}^n \{y_i - f(x_i, \mathcal{D}_n)\}^2, \quad (2.9)$$

then

$$\begin{aligned} \mathbb{E} \left[\{\hat{m}_n(x) - m(x)\}^2 \middle| \mathcal{D}_n \right] &\leq 2 \sup_{f \in \mathcal{C}_n} \left| \frac{1}{n} \sum_{i=1}^n \{y_i - f(x_i)\}^2 - \mathbb{E} \left[\{y - f(x)\}^2 \right] \right| \\ &\quad + \inf_{f \in \mathcal{C}_n} \mathbb{E} \left[\{f(x) - m(x)\}^2 \right]. \end{aligned} \quad (2.10)$$

Proof. By a similar argument, we can show that

$$\mathbb{E} \left[\{\hat{m}_n(x) - m(x)\}^2 \middle| \mathcal{D}_n \right] = \mathbb{E} \left[(y - \hat{m}_n(x))^2 \middle| \mathcal{D}_n \right] - \mathbb{E} \left[(y - m(x))^2 \right] \quad (2.11)$$

$$\begin{aligned} &= \mathbb{E} \left[(y - \hat{m}_n(x))^2 \middle| \mathcal{D}_n \right] - \inf_{f \in \mathcal{C}_n} \mathbb{E} \left[(y - f(x))^2 \right] \\ &\quad + \inf_{f \in \mathcal{C}_n} \left\{ \mathbb{E} \left[(y - f(x))^2 \right] - \mathbb{E} \left[(y - m(x))^2 \right] \right\} \end{aligned} \quad (2.12)$$

$$\begin{aligned} &= \mathbb{E} \left[(y - \hat{m}_n(x))^2 \middle| \mathcal{D}_n \right] - \inf_{f \in \mathcal{C}_n} \mathbb{E} \left[(y - f(x))^2 \right] \\ &\quad + \inf_{f \in \mathcal{C}_n} \mathbb{E} \left[(f(x) - m(x))^2 \right] \end{aligned} \quad (2.13)$$

Note we already obtained the last term in (2.10), now we continue to bound the rest:

$$\begin{aligned} &\mathbb{E} \left[(y - \hat{m}_n(x))^2 \middle| \mathcal{D}_n \right] - \inf_{f \in \mathcal{C}_n} \mathbb{E} \left[(y - f(x))^2 \right] \\ &= \sup_{f \in \mathcal{C}_n} \left\{ \mathbb{E} \left[(y - \hat{m}_n(x))^2 \middle| \mathcal{D}_n \right] - \mathbb{E} \left[(y - f(x))^2 \right] \right\} \end{aligned} \quad (2.14)$$

$$\begin{aligned} &= \sup_{f \in \mathcal{C}_n} \left(\mathbb{E} \left[(y - \hat{m}_n(x))^2 \middle| \mathcal{D}_n \right] - \frac{1}{n} \sum_{i=1}^n (y_i - \hat{m}_n(x_i))^2 + \frac{1}{n} \sum_{i=1}^n (y_i - \hat{m}_n(x_i))^2 \right. \\ &\quad \left. - \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 + \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 - \mathbb{E} \left[(y - f(x))^2 \right] \right) \end{aligned} \quad (2.15)$$

Note that by definition $\hat{m}_n := \arg \min_{f \in \mathcal{C}} f$ so it follows that

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{m}_n(x_i))^2 - \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 \leq 0 \quad (2.16)$$

and denote the right hand side of (2.15) by I , we now have

$$\begin{aligned} I &\leq \sup_{f \in \mathcal{C}_n} \left| \mathbb{E}[(y - \hat{m}_n(x))^2 | \mathcal{D}_n] - \frac{1}{n} \sum_{i=1}^n (y_i - \hat{m}_n(x_i))^2 + \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 - \mathbb{E}[(y - f(x))^2] \right| \\ &\leq \left| \mathbb{E}[(y - \hat{m}_n(x))^2 | \mathcal{D}_n] - \frac{1}{n} \sum_{i=1}^n (y_i - \hat{m}_n(x_i))^2 \right| + \sup_{f \in \mathcal{C}_n} \left| \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 - \mathbb{E}[(y - f(x))^2] \right| \\ &\leq 2 \sup_{f \in \mathcal{C}_n} \left| \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 - \mathbb{E}[(y - f(x))^2] \right|. \end{aligned} \quad (2.17)$$

■

2.3 Linear Projection Coefficient

We may wonder which class $\mathcal{C}_n$ to choose from? If $\mathcal{C}_n$ is too large, overfitting may occur and the estimation error will likely increase; however if $\mathcal{C}_n$ is too small the estimation error may decrease but the approximation error is likely to increase. We will now focus on the class $\mathbb{L}$ of linear functions:

$$\mathbb{L} : \left\{ f : \mathbb{R}^p \rightarrow \mathbb{R}, \forall x \in \mathbb{R}^p, \exists \beta \in \mathbb{R}^p, f(x) = x^\top \beta \right\} \quad (2.18)$$

Definition

Definition 3. The best linear predictor of y given x is, if it exists, the function I^* satisfying

$$I^* = \arg \min_{f \in \mathbb{L}} \mathbb{E}[\{y - f(x)\}^2]. \quad (2.19)$$

Since $I^* \in \mathbb{L}$, there exists $\beta \in \mathbb{R}^p$ such that $I^*(x) = x^\top \beta$, the vector β is called the **linear projection coefficient**. Note that if we have a function $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^2$, then it is not linear if the covariates are chosen as $x = (x_1, x_2)^\top$. But it would be linear if we choose the covariates to be $x = (x_1, x_2, x_2^2)^\top$. So it is important to specify our covariates in advance, and we do not change them during the study.

Theorem

Theorem 11. Under regularity conditions:

$$(H1) \mathbb{E}[y^2] < \infty; (H2) \mathbb{E}[||x||_2^2] < \infty; (H3) \mathbb{E}[xx^\top] \text{ is positive definite.}$$

Then the best linear predictor of y given x , denoted I^* exists and is unique, and is fully determined by the unique linear projection coefficient

$$\beta := \left(\mathbb{E}[xx^\top] \right)^{-1} \mathbb{E}[xy]. \quad (2.20)$$

We first provide some technical lemma in convex optimization. We first say a set $\mathcal{C} \subseteq \mathbb{R}^p$ is **convex**, if $\forall \mathbf{x}, \mathbf{y} \in \mathcal{C}, t \in [0, 1]$, we have $t\mathbf{x} + (1 - t)\mathbf{y} \in \mathcal{C}$.

Definition

Definition 4. A function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ defined on a convex domain is convex, if for all $\mathbf{x}, \mathbf{y} \in \mathcal{C}, t \in [0, 1]$,

$$f(t\mathbf{x} + (1 - t)\mathbf{y}) \leq tf(\mathbf{x}) + (1 - t)f(\mathbf{y}). \quad (2.21)$$

If strict inequality holds then we say f is strictly convex, f is strictly concave if $-f$ is strictly convex.

The theory we will use is that: if f is convex, then any local minimizer is also a global minimizer, if f is strictly convex, then the minimizer (if exists) is unique. Its gradient can be found by Taylor expansion:

$$f(\mathbf{u} + \mathbf{h}) = f(\mathbf{u}) + \langle \mathbf{h}, \nabla f(\mathbf{u}) \rangle + o(\|\mathbf{h}\|_2). \quad (2.22)$$

Hence we can easily find the gradient in this way, and solve for $\nabla f(\mathbf{u}) = 0$. The sketch of the proof will be: First show $F : \mathbb{R}^p \rightarrow \mathbb{R}, \mathbf{u} \rightarrow \mathbb{E}[(\mathbf{y} - \mathbf{x}^\top \mathbf{u})^2]$ is strictly convex and finite, then solve $\nabla F(\mathbf{u}) = 0$ and in this case $\mathbf{u}$ will be the arg min by the theory we just developed.

We first show β in the above theorem is indeed the minimizer.

Proof. Note that we have

$$F(\mathbf{u}) = \underbrace{\mathbb{E}[y^2]}_{F_1(\mathbf{u})} - \underbrace{2\mathbf{u}^\top \mathbb{E}[y\mathbf{x}]}_{F_2(\mathbf{u})} + \underbrace{\mathbf{u}^\top \mathbb{E}[\mathbf{x}\mathbf{x}^\top] \mathbf{u}}_{F_3(\mathbf{u})} \quad (2.23)$$

We later will show F is indeed finite and convex, but now it suffices to find the gradient for each term first and compute the minimizer. In F_1 , since it is independent of $\mathbf{u}$ so the answer is simply zero. In F_2 , we have

$$F_2(\mathbf{u} + \mathbf{h}) = -2\mathbb{E}[y\mathbf{x}^\top](\mathbf{u} + \mathbf{h}) \quad (2.24)$$

$$= -2\mathbb{E}[y\mathbf{x}^\top]\mathbf{u} - 2\mathbb{E}[y\mathbf{x}^\top]\mathbf{h} \quad (2.25)$$

$$= F_2(\mathbf{u}) + \mathbf{h}^\top (-2\mathbb{E}[y\mathbf{x}]) \quad (2.26)$$

where $\nabla F_2(\mathbf{u}) = -2\mathbb{E}[y\mathbf{x}]$, and similarly

$$F_3(\mathbf{u} + \mathbf{h}) = (\mathbf{u} + \mathbf{h})^\top (\mathbf{u} + \mathbf{h}) \mathbb{E}[\mathbf{x}\mathbf{x}^\top] \quad (2.27)$$

$$= (\mathbf{u}^\top \mathbf{u} + 2\mathbf{u}^\top \mathbf{h} + \mathbf{h}^\top \mathbf{h}) \mathbb{E}[\mathbf{x}\mathbf{x}^\top] \quad (2.28)$$

$$= \mathbf{u}^\top \mathbb{E}[\mathbf{x}\mathbf{x}^\top] \mathbf{u} + 2\mathbf{u}^\top \mathbb{E}[\mathbf{x}\mathbf{x}^\top] \mathbf{h} + o(\|\mathbf{h}\|_2) \quad (2.29)$$

and hence $\nabla F_3(\mathbf{u}) = 2\mathbf{u} \mathbb{E}[\mathbf{x}\mathbf{x}^\top]$, and finally we solve $\nabla F = 0$, that is,

$$-2\mathbb{E}[y\mathbf{x}] + 2\mathbb{E}[\mathbf{x}\mathbf{x}^\top]\mathbf{u} = 0. \quad (2.30)$$

By assumption $\mathbb{E}[\mathbf{x}\mathbf{x}^\top]$ is positive definite, so it is invertible and hence we have

$$\mathbf{u} = \mathbb{E}[\mathbf{x}\mathbf{x}^\top]^{-1} \cdot \mathbb{E}[y\mathbf{x}]. \quad (2.31)$$

■

We then show F is strictly convex and finite:

Proof. Note that, the function

$$F(\mathbf{u}) = \mathbb{E}[(y - \mathbf{x}^\top \mathbf{u})^2] \quad (2.32)$$

can be written as

$$F(\mathbf{u}) = \underbrace{\mathbb{E}[y^2]}_{F_1(\mathbf{u})} - \underbrace{2\mathbf{u}^\top \mathbb{E}[y\mathbf{x}^\top]}_{F_2(\mathbf{u})} + \underbrace{\mathbf{u}^\top \mathbf{u} \mathbb{E}[\mathbf{x}\mathbf{x}^\top]}_{F_3(\mathbf{u})} \quad (2.33)$$

where by assumption the first term is already finite. In the second term, note that by Jensen's inequality we have

$$\|\mathbb{E}[y\mathbf{x}]\|_2 \leq \mathbb{E}[\|y\mathbf{x}\|_2] = \mathbb{E}[|y| \cdot \|\mathbf{x}\|_2] \quad (2.34)$$

and by Cauchy-Schwarz inequality, where the inner product is defined by $\langle Z_1, Z_2 \rangle = \mathbb{E}[Z_1 Z_2]$, we have

$$\mathbb{E}[|y| \cdot \|\mathbf{x}\|_2] \leq \mathbb{E}[|y|^2]^{1/2} \cdot \mathbb{E}[\|\mathbf{x}\|_2^2]^{1/2} \quad (2.35)$$

which is finite by assumptions. Finally for the last term we again use Jensen's inequality,

$$\mathbf{u}^\top \mathbb{E}[\mathbf{x}\mathbf{x}^\top] \mathbf{u} \leq \|\mathbf{u}\|_2^2 \cdot \|\mathbb{E}[\mathbf{x}\mathbf{x}^\top]\|_2 \leq \mathbb{E}[\|\mathbf{x}\mathbf{x}^\top\|_2] = \mathbb{E}[\|\mathbf{x}\|_2^2] < \infty \quad (2.36)$$

by assumption. Finally, $\forall \theta \in (0, 1)$, we have

$$F(\theta \mathbf{u} + (1 - \theta) \mathbf{v}) = \mathbb{E}[(y - \theta \mathbf{x}^\top \mathbf{u} - (1 - \theta) \mathbf{x}^\top \mathbf{v})^2] \quad (2.37)$$

$$= \mathbb{E}[(\theta y + (1 - \theta) y - \theta \mathbf{x}^\top \mathbf{u} - (1 - \theta) \mathbf{x}^\top \mathbf{v})^2] \quad (2.38)$$

$$= \mathbb{E}[(\theta(y - \mathbf{x}^\top \mathbf{u}) + (1 - \theta)(y - \mathbf{x}^\top \mathbf{v}))^2] \quad (2.39)$$

$$\leq \mathbb{E}[\theta(y - \mathbf{x}^\top \mathbf{u})^2 + (1 - \theta)(y - \mathbf{x}^\top \mathbf{v})^2] \quad (2.40)$$

$$= \theta \mathbb{E}[(y - \mathbf{x}^\top \mathbf{u})^2] + (1 - \theta) \mathbb{E}[(y - \mathbf{x}^\top \mathbf{v})^2]. \quad (2.41)$$

■

REMARKS

1. We assumed that $\mathbf{x} \in \mathbb{R}^p$ so in this case $\mathbf{x}^\top \mathbf{x} \in \mathbb{R}$ is a scalar and $\mathbf{x}\mathbf{x}^\top$ is a $p \times p$ matrix. The matrix $\mathbb{E}[\mathbf{X}\mathbf{X}^\top]$ is also known as the random matrix defined by

$$\mathbb{E}[\mathbf{X}\mathbf{X}^\top] = \begin{pmatrix} \mathbb{E}[X_{11}] & \cdots & \mathbb{E}[X_{1n}] \\ \vdots & \ddots & \vdots \\ \mathbb{E}[X_{n1}] & \cdots & \mathbb{E}[X_{nn}] \end{pmatrix} \quad (2.42)$$

2. *Jensen's inequality:* Let $f(x)$ be convex, then $f(\mathbb{E}[X]) \leq \mathbb{E}[f(x)]$, and if f is concave we have $\geq$ sign. When f is strictly convex/concave, we drop the equal sign.

3. *Cauchy-Schwarz inequality:* Let $\langle \cdot, \cdot \rangle$ be an inner product defined on the inner product space V . Then $\forall x, y \in V$, $|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$.

Proposition

Proposition 3. *With the same assumption in theorem 11, the linear projection coefficient*

$$\beta = \left(\mathbb{E}[xx^\top] \right)^{-1} \cdot \mathbb{E}[xy] \quad (2.43)$$

also solves

$$\beta = \arg \min_{u \in \mathbb{R}^p} \mathbb{E} \left[\left(m(x) - x^\top u \right)^2 \right] \quad (2.44)$$

where $m(x) = \mathbb{E}[y|x]$.

Proof. The proof is straightforward. We need to know in fact $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|X]]$, then

$$\mathbb{E} \left[\left(m(x) - x^\top u \right)^2 \right] = \mathbb{E} \left[m(x)^2 - 2m(x)x^\top u + (x^\top u)^2 \right] \quad (2.45)$$

$$= \mathbb{E} \left[\mathbb{E}^2[y|x] - 2\mathbb{E}[y|x]x^\top u + (x^\top u)^2 \right] \quad (2.46)$$

$$= \mathbb{E}[\mathbb{E}^2[y|x]] - 2\mathbb{E}[\mathbb{E}[y|x]x^\top u] + \mathbb{E}[(x^\top u)^2] \quad (2.47)$$

$$= \mathbb{E}[y^2] - \mathbb{E}[\mathbb{V}(y|x)] - 2\mathbb{E}[yx^\top u] + \mathbb{E}[(x^\top u)^2] \quad (2.48)$$

$$= \mathbb{E}[(y - x^\top u)^2] - \mathbb{E}[\mathbb{V}(y|x)] \quad (2.49)$$

where the last term is independent of u . ■

With our linear coefficient $x^\top \beta$, we recall previously we defined

$$y_i = m(x_i) + \varepsilon_i, \quad i = 1, 2, \dots, n \quad (2.50)$$

we now can write

$$y_i = x_i^\top \beta + e_i, \quad i = 1, 2, \dots, n \quad (2.51)$$

where $e_i = y_i - x_i^\top \beta$, and it is denoted as the “noise term”. (Note that our notation is kinda unclear, x_i is the i th sample, not covariates, since $i = [n]$ and $j = [p]$ for covariates.)

Proposition

Proposition 4. *For $i = 1, 2, \dots, n$, $\mathbb{E}[x_i e_i] = \mathbf{0} \in \mathbb{R}^p$.*

Proof. By direct computation, we have

$$\mathbb{E}[x_i e_i] = \mathbb{E}[x_i (y_i - x_i^\top \beta)] \quad (2.52)$$

$$= \mathbb{E}[x_i y_i - x_i x_i^\top \beta] \quad (2.53)$$

$$= \mathbb{E}[x_i y_i] - \mathbb{E}[x_i x_i^\top] \beta \quad (2.54)$$

$$= \mathbb{E}[x_i y_i] - \mathbb{E}[x_i x_i^\top] \mathbb{E}(x_i x_i^\top)^{-1} \mathbb{E}[x_i y_i] \quad (2.55)$$

$$= \mathbb{E}[x_i y_i] - \mathbf{1}_p \mathbb{E}[x_i y_i] \quad (2.56)$$

$$= \mathbf{0}_p. \quad (2.57)$$

■

Definition

Definition 5. An intercept is any constant covariate x_j , i.e, there exists $C \in \mathbb{R}$ such that $x_j = C$ almost surely.

From the previous proposition, we can see that

$$\mathbb{E}[x_i e_i] = \begin{pmatrix} \mathbb{E}[x_{i1} e_i] \\ \mathbb{E}[x_{i2} e_i] \\ \vdots \\ \mathbb{E}[x_{ip} e_i] \end{pmatrix} = \mathbf{0}_p \quad (2.58)$$

while if $x_{ij} = C$ for some covariates, then we can see $\mathbb{E}[e_i] = 0$ must be true. Usually, with out loss of generality the intercept is usually taken to be equal to 1 in the first position. Now consider another decomposition:

$$y = \mathbf{x}^\top \boldsymbol{\beta} + \underbrace{(m(\mathbf{x}) - \mathbf{x}^\top \boldsymbol{\beta})}_{\eta(\mathbf{x})} + \underbrace{(y - m(\mathbf{x}))}_{\varepsilon} \quad (2.59)$$

where $m(\mathbf{x}) := \mathbb{E}[y|\mathbf{x}]$, which takes the form $y = \mathbf{x}^\top \boldsymbol{\beta} + \eta(\mathbf{x}) + \varepsilon$, we then study some properties of the form (2.59) with intercept introduced:

Proposition

Proposition 5. (Weak Orthogonality) Let $x_j \in [p]$ be covariates, then $\mathbb{E}[\eta(\mathbf{x})x_j] = 0$ for all $j = [p]$.

Proof. Previously we have shown that $\mathbb{E}[\mathbf{x}(y - \mathbf{x}^\top \boldsymbol{\beta})] = \mathbf{0}_p$, so it must also hold for x_j , so $\mathbb{E}[x_j(y - \mathbf{x}^\top \boldsymbol{\beta})] = 0$. Note that $y - \mathbf{x}^\top \boldsymbol{\beta} = \eta(\mathbf{x}) + \varepsilon$, so we have

$$\mathbb{E}[x_j(y - \mathbf{x}^\top \boldsymbol{\beta})] = \mathbb{E}[x_j \eta(\mathbf{x})] + \mathbb{E}[x_j \varepsilon] = 0, \quad (2.60)$$

where

$$\mathbb{E}[x_j \varepsilon] = \mathbb{E}[\mathbb{E}[x_j \varepsilon | \mathbf{x}]] = \mathbb{E}[x_j \mathbb{E}[\varepsilon | \mathbf{x}]] = \mathbb{E}[x_j \cdot 0] = 0, \quad (2.61)$$

which implies $\mathbb{E}[\eta(\mathbf{x})x_j] = 0$ for all $j = [p]$. ■

Proposition

Proposition 6. (Strong Orthogonality) For all $f \in L^2(\mathbb{P}_x)$, $\mathbb{E}[f(\mathbf{x})\varepsilon] = 0$.

Proof. From (2.59), we see that $\varepsilon = y - \mathbf{x}^\top \boldsymbol{\beta} - m(\mathbf{x}) + \mathbf{x}^\top \boldsymbol{\beta} = y - m(\mathbf{x})$, so we have

$$\mathbb{E}[f(\mathbf{x})\varepsilon] = \mathbb{E}[f(\mathbf{x})(y - m(\mathbf{x}))], \quad (2.62)$$

also note that

$$\mathbb{E}[f(\mathbf{x})(y - m(\mathbf{x}))] = \mathbb{E}[\mathbb{E}[f(\mathbf{x})(y - m(\mathbf{x})) | \mathbf{x}]] = \mathbb{E}[f(\mathbf{x})(\mathbb{E}[y | \mathbf{x}] - m(\mathbf{x}))] \equiv 0 \quad (2.63)$$

where $m(\mathbf{x}) := \mathbb{E}[y | \mathbf{x}]$. ■

Proposition

Proposition 7. Suppose intercept is included among the covariates, then in the decomposition

$$y = \mathbf{x}^\top \boldsymbol{\beta} + \underbrace{(m(\mathbf{x}) - \mathbf{x}^\top \boldsymbol{\beta})}_{\eta(\mathbf{x})} + \underbrace{(y - m(\mathbf{x}))}_{\varepsilon}, \quad (2.64)$$

we have $\mathbb{E}[\eta(\mathbf{x})] = 0$ and $\mathbb{E}[\varepsilon\eta(\mathbf{x})] = 0$.

Proof. As we have shown before, if intercept is included then $\mathbb{E}[e] = 0$ when $y = \mathbf{x}^\top \boldsymbol{\beta} + e$, in this case we have $e = m(\mathbf{x}) - \mathbf{x}^\top \boldsymbol{\beta} + y - m(\mathbf{x}) = y - \mathbf{x}^\top \boldsymbol{\beta}$ so $\mathbb{E}[y - \mathbf{x}^\top \boldsymbol{\beta}] = 0$, then by a conditional argument we have

$$\mathbb{E}[y - \mathbf{x}^\top \boldsymbol{\beta}] = \mathbb{E}[\mathbb{E}[y|\mathbf{x}] - \mathbf{x}^\top \boldsymbol{\beta}] := \mathbb{E}[\eta(\mathbf{x})] = 0. \quad (m(\mathbf{x}) := \mathbb{E}[y|\mathbf{x}]) \quad (2.65)$$

As for the second statement, we have

$$\mathbb{E}[\varepsilon\eta(\mathbf{x})] = \mathbb{E}[\mathbb{E}[\varepsilon\eta(\mathbf{x})|\mathbf{x}]] = \mathbb{E}[\eta(\mathbf{x}) \cdot \mathbb{E}[\varepsilon|\mathbf{x}]] = \mathbb{E}[0 \cdot 0] \equiv 0. \quad (2.66)$$

■

REMARKS

We may think of expectation as an inner product. Consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ we define $L^2 := L^2(\Omega, \mathcal{F}, \mathbb{P})$ as the space of square-integrable random variables with

$$\mathbb{E}[X^2] = \int_{\Omega} X(\omega)^2 \mathbb{P}(d\omega) < \infty. \quad (2.67)$$

We claim that L^2 defines a Hilbert space (a complete inner product space). Then the inner product is defined by $\langle X, Y \rangle := \mathbb{E}[XY]$ and $\langle X, Y \rangle = 0$ if and only if X, Y are independent (or called orthogonal). The norm equipped on L^2 is simply defined by $\|X\|_{L^2} := \mathbb{E}[X^2]^{1/2}$.

In a Hilbert space F , let $E \subseteq F$ closed, then the projection theorem says that $\forall x \in F$, there is a unique element $P_E(x) \in E$ such that

$$\|x - P_E(x)\| = \inf_{y \in E} \|x - y\| \quad (2.68)$$

if $h \in E$, $\langle x - h, y \rangle = 0$ for every $y \in E$, then $h = P_E(x)$, denote by the orthogonal projection of x onto E . Based on that, we know every $x \in F$ has a unique decomposition $x = P_E(x) + P_{E^\perp}(x)$ where $P_E(x) \in E, P_{E^\perp}(x) \in E^\perp$. The mapping $\mathcal{P} : x \mapsto P_E(x)$ is called the orthogonal projection operator.

Then we can use projection argument to prove the above propositions easily. Note that to show $\mathbb{E}[x\varepsilon] = 0$, recall our definition that $y = \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon$, it can also be viewed as a projection:

$$y = \mathcal{P}_{\mathcal{L}}(y) + \mathcal{P}_{\mathcal{L}^\perp}(y) \quad (2.69)$$

where $\mathcal{L}$ denotes the space of linear functions, and we can see that $\mathbf{x}^\top \boldsymbol{\beta} \in \mathcal{L}, \varepsilon \in \mathcal{L}^\perp$ and we have

$$\mathbb{E}[\mathbf{x}\varepsilon] = \langle \mathbf{x}, \varepsilon \rangle = \mathbf{0}_p. \quad (2.70)$$

And similarly for $\mathbb{E}[f(\mathbf{x})\varepsilon]$ we have

$$\mathbf{y} = \mathcal{P}_{\mathcal{G}}(\mathbf{y}) + \mathcal{P}_{\mathcal{G}^\perp}(\mathbf{y}) \quad (2.71)$$

where $\mathcal{G}$ denotes the space of functions $f \in L^2(\mathbb{P}_{\mathbf{x}})$ and hence $\mathbb{E}[f(\mathbf{x})\varepsilon] = 0$.

2.4 Design Matrix and Linear Models

Further, we may present the data $(\mathbf{x}_i, y_i)$ in a matrix form:

Definition

Definition 6. The covariates are stored in the design matrix $\mathbf{X} \in \mathcal{M}_{n,p}$ where X_{ij} denotes the value of covariate j for observation $i \in [n]$, we have

$$\mathbf{X} = \begin{pmatrix} 1 & X_{1,2} & \cdots & X_{1,p} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ 1 & X_{n,2} & \cdots & X_{n,p} \end{pmatrix} \quad (2.72)$$

where we assume intercept is included, and we put it as the first covariate. The response is represented by a vector $\mathbf{y} = (y_1, \dots, y_n)^\top$.

The design matrix $\mathbf{X}$ is said to be **full rank** if $\mathbb{P}[\text{rank}(\mathbf{X}) = p] = 1$. The covariates are said to be **multicollinear** if it is not full rank, and in this case we may discard some of the covariates since they are considered as “redundant” information”. The good thing about full rank is that, if $\mathbf{X}$ is full rank, then $\mathbf{X}^\top \mathbf{X}$ will be invertible, which can make our life easier later. In the setting of a random design, $\mathbf{X}$ is random; while in a fixed design $\mathbf{X}$ is deterministic, and the intercept column corresponds to a degenerate distribution at 1.

Definition

Definition 7. We denote $\mathcal{D}_n = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ to be the i.i.d data set we get, we say the data follow a linear model if $\mathbb{E}[y|\mathbf{x}] = \mathbf{x}^\top \boldsymbol{\beta}$, i.e, $m(\mathbf{x}) \in \mathbb{L}$. The model is homoscedastic if $\mathbb{V}(y|\mathbf{x}) = \mathbb{V}(\varepsilon|\mathbf{x}) = \sigma^2$ a constant, and the model is heteroscedastic if it is not a constant.

Proposition

Proposition 8. If $\mathbb{P}_{\mathbf{x}}$ has a continuous distribution with respect to the Lebesgue measure on $\mathbb{R}^p$, then the design matrix $\mathbf{X}$ is full rank.

Proof. ■

We now introduce two different regression models, the **jointly Gaussian linear model** and **normal linear model**:

Jointly Gaussian linear model: This model is often viewed as the main motivation for the full-rank, homoscedastic linear model. We assume $(\mathbf{x}, y)^\top$ is jointly Gaussian in the linear model

$$y = \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon \quad (2.73)$$

where we have

$$\text{Cov}(\mathbf{x}, \varepsilon) = \mathbb{E}[\mathbf{x}\varepsilon] - \mathbb{E}[\mathbf{x}] \cdot \mathbb{E}[\varepsilon] = \mathbf{0}_p. \quad (2.74)$$

This is because by definition $\mathbb{E}[\varepsilon] = \mathbb{E}[\mathbb{E}[\varepsilon|\mathbf{x}]] = \mathbb{E}[0] = 0$ and furthermore

$$\mathbb{E}[\mathbf{x}\varepsilon] = \mathbb{E}[\mathbb{E}[\mathbf{x}\varepsilon|\mathbf{x}]] = \mathbb{E}[\mathbf{x}\mathbb{E}[\varepsilon|\mathbf{x}]] = \mathbb{E}[\mathbf{x} \cdot 0] = \mathbf{0}. \quad (2.75)$$

In a jointly Gaussian model, once we have the covariance is zero, we may conclude that $\mathbf{x}, \varepsilon$ are independent, and since Gaussianity is preserved under linear transformations, it follows that $(\mathbf{x}, \varepsilon)^\top$ is also a jointly Gaussian, and by independence, $\mathbb{V}(\varepsilon|\mathbf{x}) = \mathbb{V}(\varepsilon)$ which means the model is homoscedastic.

Normal Linear Model: This is a relaxation of the jointly Gaussian model, it assumes that the residuals are Gaussian, and the covariates are independent of the residuals. It is less restrictive than the jointly Gaussian model because here the covariates may follow any distribution, and normal linear model is therefore still a particular case of a homoscedastic linear model.

In the case of a linear model, one of the main focuses is to make inference about the unknown vector of coefficients $\boldsymbol{\beta}$.

Definition

Definition 8. A statistical model is a class of distributions $\{\mathbb{P}_\theta; \theta \in \Theta\}$ that we believe might have generated the data $\mathcal{D}_n$, the set Θ is called the parameter space and its elements are the parameters of the model.

Parameters can often be partitioned as $\boldsymbol{\theta} = [\boldsymbol{\gamma}^\top, \boldsymbol{\eta}^\top]^\top$ where our primary interest is called **parameter of interest** $\boldsymbol{\gamma}$ and the remaining component $\boldsymbol{\eta}$ is not of direct interest, but is still required to fully specify the distribution and it called a **nuisance parameter**. There are three models:

- (i) If the parameter space is finite dimensional, the model is called **parametric**;
- (ii) If the parameter of interest lies in a finite dimensional space, but there exists a nuisance parameter described by an infinite dimensional space, the model is called **semi-parametric**;
- (iii) If the parameter of interest cannot be indexed by a finite dimensional space, the model is called **non-parametric**.

Proposition

Proposition 9. *A random design linear model is semi-parametric; A fixed design linear model with Gaussian residuals is a parametric model.*

2.5 Closing Remark

Linear regression analysis may have several objectives, for example: (i) Estimate the regression function $m(\mathbf{x})$; (ii) Make predictions on new/unobserved points, especially in random design; (iii) Recover the true signal $m(\mathbf{x}_i)$ from noisy observations $y_i = m(\mathbf{x}_i) + \varepsilon_i$, and this is sometimes called denoising; (iv) Understand how a covariate $\mathbf{X}_j$ affects the response y .

However, there are limitations of traditional linear models: (i) The regression function may not be linear at all; (ii) The data may exhibit dependence; (iii) The data may not be identically distributed; (iv) The conditional variance may not be constant; (v) The data may be high-dimensional where $p \gg n$.

That motivates us to continue our study on regression.

The Ordinary Least Squares Estimator

3.1 Derivations of the OLS Estimator

Previously, we have shown that if we assume a linear model, then the regression function can be written as

$$m(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta} = \sum_{j=1}^p x_j \beta_j \quad (3.1)$$

and the problem of estimating the regression function reduces to the problem of estimating $\boldsymbol{\beta}$, given by $\boldsymbol{\beta} = \mathbb{E}[\mathbf{x}\mathbf{x}^\top]^{-1}\mathbb{E}[\mathbf{x}y]$. However in reality we do not have access to the true expectation $\mathbb{E}[\cdot]$, what we normally do is to replace $\mathbb{E}[\cdot]$ by its empirical estimate

$$\hat{\mathbb{E}} = \frac{1}{n} \sum_{i \in [n]} (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2. \quad (3.2)$$

Definition

Definition 9. The ordinary least squares loss over $\mathcal{D}_n$ is the function $\mathcal{O}_{LS}(\cdot, \mathcal{D}_n) := \mathcal{O}_{LS,n}$ from $\mathbb{R}^p$ to $\mathbb{R}$ defined by

$$\mathcal{O}_{LS,n}(\boldsymbol{\beta}) := \frac{1}{n} \sum_{i \in [n]} \{y_i - \mathbf{x}_i^\top \boldsymbol{\beta}\}^2 := \frac{1}{n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2. \quad (3.3)$$

An ordinary least squares estimator of $\boldsymbol{\beta}$ is any vector $\hat{\boldsymbol{\beta}}_{ols}(\mathcal{D}_n) := \hat{\boldsymbol{\beta}}_{ols,n}$ satisfying

$$\hat{\boldsymbol{\beta}}_{ols,n} \in \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \mathcal{O}_{LS,n}(\boldsymbol{\beta}). \quad (3.4)$$

Note that sometimes the residual sum of squares (RSS) is considered to be the loss of reference, where

$$RSS(\boldsymbol{\beta}) = \sum_{i \in [n]} \{y_i - \mathbf{x}_i^\top \boldsymbol{\beta}\}^2 = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = n \cdot \mathcal{O}_{LS,n}. \quad (3.5)$$

but they are considered equivalent since are up to some constants.

Theorem

Theorem 12. The ordinary least squares estimator $\hat{\boldsymbol{\beta}}_n$ of $\boldsymbol{\beta}$ satisfies the normal equations

$$(\mathbf{X}^\top \mathbf{X}) \hat{\boldsymbol{\beta}}_n := \mathbf{X}^\top \mathbf{y}. \quad (3.6)$$

Additionally, if the design matrix is full rank, then the ordinary least squares estimator $\hat{\boldsymbol{\beta}}_{ols,n} := \hat{\boldsymbol{\beta}}_n$ of $\boldsymbol{\beta}$ exists, and it is unique, given by

$$\hat{\boldsymbol{\beta}}_n := (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \left(\sum_{i \in [n]} \mathbf{x}_i \mathbf{x}_i^\top \right)^{-1} \cdot \sum_{i \in [n]} \mathbf{x}_i y_i. \quad (3.7)$$

Proof. The proof is similar to **theorem 11**, where we will use the fact that the function $\mathcal{O}_{LS,n}(\boldsymbol{\beta})$ is strictly convex, and it hence suffices to compute the gradient and equate to zero to solve for $\boldsymbol{\beta}$. Let

$$F(\mathbf{u}) = \frac{1}{n} \sum_{i \in [n]} \{y_i - \mathbf{x}_i^\top \mathbf{u}\}^2 \quad (3.8)$$

$$= \frac{1}{n} \sum_{i=1}^n (y_i^2 - 2y_i \mathbf{x}_i^\top \mathbf{u} + \mathbf{u}^\top \mathbf{x}_i \mathbf{x}_i^\top \mathbf{u}) \quad (3.9)$$

$$= \underbrace{\frac{1}{n} \sum_{i=1}^n y_i^2}_{F_1(\mathbf{u})} - \underbrace{\frac{2}{n} \sum_{i=1}^n y_i \mathbf{x}_i^\top \mathbf{u}}_{F_2(\mathbf{u})} + \underbrace{\frac{1}{n} \mathbf{u}^\top \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \mathbf{u}}_{F_3(\mathbf{u})} \quad (3.10)$$

where we compute the gradient separately. Note that $F_1(\mathbf{u})$ is independent of $\mathbf{u}$ so we simply have $\nabla F_1(\mathbf{u}) = 0$. Then since the function F is strictly convex, we just need to find $\mathbf{u}^*$ that cancels out the gradient and $\mathbf{u}^*$ is the unique global minimum. We have

$$F_2(\mathbf{u} + \mathbf{h}) = -\frac{2}{n} \sum_{i=1}^n y_i \mathbf{x}_i^\top (\mathbf{u} + \mathbf{h}) \quad (3.11)$$

$$= -\frac{2}{n} \sum_{i=1}^n y_i \mathbf{x}_i^\top \mathbf{u} - \mathbf{h}^\top \frac{2}{n} \sum_{i=1}^n y_i \mathbf{x}_i \quad (3.12)$$

$$:= F_2(\mathbf{u}) + \mathbf{h}^\top \nabla F_2(\mathbf{u}), \quad (3.13)$$

and we have

$$\nabla F_2(\mathbf{u}) = -\frac{2}{n} \sum_{i=1}^n y_i \mathbf{x}_i. \quad (3.14)$$

Similarly for $F_3(\mathbf{u})$:

$$F_3(\mathbf{u} + \mathbf{h}) = (\mathbf{u} + \mathbf{h})^\top \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top (\mathbf{u} + \mathbf{h}) \quad (3.15)$$

$$= \mathbf{u}^\top \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \mathbf{u} + \mathbf{h}^\top \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \mathbf{u} + \mathbf{u}^\top \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \mathbf{h} + \mathbf{h}^\top \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \mathbf{h} \quad (3.16)$$

$$= F_3(\mathbf{u}) + \mathcal{O}(\|\mathbf{h}\|_2) + \frac{2}{n} \mathbf{h}^\top \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \mathbf{u} \quad (3.17)$$

and again we the have

$$\nabla F_3(\mathbf{u}) = \frac{2}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \mathbf{u}. \quad (3.18)$$

We finally set $\nabla F(\mathbf{u}) = \mathbf{0}_p$ and we have

$$-\frac{2}{n} \sum_{i=1}^n y_i \mathbf{x}_i + \frac{2}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \mathbf{u} = \mathbf{0}_p \quad (3.19)$$

we solve for $\mathbf{u}$ and we get

$$\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \mathbf{u} = \sum_{i=1}^n y_i \mathbf{x}_i \quad (3.20)$$

which is equivalent as $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X} \mathbf{u}$. Further if $\mathbf{X}^\top \mathbf{X}$ is invertible, then we can solve $\mathbf{u}$ as

$$\mathbf{u} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (3.21)$$

■

REMARK

We have used several properties from linear algebra: Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^p$, then (i) $\mathbf{a}^\top \mathbf{b} = \mathbf{b}^\top \mathbf{a}$; (ii) $\mathbf{a} \mathbf{a}^\top \in M_{p \times p}(\mathbb{R})$ is symmetric; (iii) $(\mathbf{A} \mathbf{B} \mathbf{C})^\top = \mathbf{C}^\top \mathbf{B}^\top \mathbf{A}^\top$ for matrices.

We may implement **theorem 12** to a simple linear model:

Proposition

Proposition 10. Suppose we have observable data $\mathcal{D} := \{(x_1, y_1), \dots, (x_n, y_n)\}$ ($(x_i, y_i) \in \mathbb{R}^2$) and suppose we have a regression model $y = \beta_0 + \beta_1 x + \varepsilon$ with $\beta_0, \beta_1 \in \mathbb{R}$ as parameters, then

$$\hat{\beta}_0 = \bar{y}_n - \bar{x}_n \hat{\beta}_1 \quad \hat{\beta}_1 = \frac{\sum_{i \in [n]} (x_i - \bar{x}_n)(y_i - \bar{y}_n)}{\sum_{i \in [n]} (x_i - \bar{x}_n)^2} \quad (3.22)$$

Proof. Note that we have

$$\mathcal{O}_{LS}(\beta_1, \beta_2) = \sum_{i=1}^n \{y_i - \mathbf{x}_i^\top \boldsymbol{\beta}\}^2 \quad (3.23)$$

$$= \arg \min_{\beta_0, \beta_1} \frac{1}{n} \sum_{i=1}^n \{y_i - (\beta_0 + \beta_1 x_i)\}^2 \quad (3.24)$$

$$= \arg \min_{\beta_0, \beta_1} \frac{1}{n} \sum_{i=1}^n \{y_i^2 + \beta_0^2 + 2\beta_0\beta_1 x_i + \beta_1^2 x_i^2 - 2y_i\beta_0 - 2\beta_1 y_i x_i\} \quad (3.25)$$

Further, y_i^2 is independent of β_0, β_1 , and we use the fact that

$$\frac{1}{n} \sum_{i=1}^n x_i := \bar{x}_n \quad \frac{1}{n} \sum_{i=1}^n y_i := \bar{y}_n \quad (3.26)$$

to obtain

$$\mathcal{O}_{LS}(\beta_1, \beta_2) := \arg \min_{\beta_0, \beta_1} \left\{ \beta_0^2 + \frac{1}{n} \sum_{i=1}^n x_i^2 \beta_1^2 + 2\bar{x}_n \beta_0 \beta_1 - 2\bar{y}_n \beta_0 - \frac{2}{n} \sum_{i=1}^n x_i y_i \beta_1 \right\}. \quad (3.27)$$

Define a multivariate function $f(\beta_0, \beta_1)$ by

$$f(\beta_0, \beta_1) = \beta_0^2 + \frac{1}{n} \sum_{i=1}^n x_i^2 \beta_1^2 + 2\bar{x}_n \beta_0 \beta_1 - 2\bar{y}_n \beta_0 - \frac{2}{n} \sum_{i=1}^n x_i y_i \beta_1 \quad (3.28)$$

and the gradient of f is given by

$$\nabla f = \left(\frac{\partial f}{\partial \beta_0}, \frac{\partial f}{\partial \beta_1} \right) = \left(2\beta_0 + 2\bar{x}_n \beta_1 - 2\bar{y}_n, \frac{2}{n} \sum_{i=1}^n x_i^2 \beta_1 + 2\bar{x}_n \beta_0 - \frac{2}{n} \sum_{i=1}^n x_i y_i \right). \quad (3.29)$$

Set $\nabla f = 0$, we solve the system

$$\begin{cases} 2\beta_0 + 2\bar{x}_n \beta_1 - 2\bar{y}_n & = 0 \\ \frac{2}{n} \sum_{i=1}^n x_i^2 \beta_1 + 2\bar{x}_n \beta_0 - \frac{2}{n} \sum_{i=1}^n x_i y_i & = 0 \end{cases} \quad (3.30)$$

where the first equation will directly yield

$$\hat{\beta}_0 = \bar{y}_n - \bar{x}_n \hat{\beta}_1. \quad (3.31)$$

Now use (3.31) to solve for $\hat{\beta}_1$ in the second equation, we have

$$\frac{1}{n} \sum_{i=1}^n x_i^2 \hat{\beta}_1 + \bar{x}_n (\bar{y}_n - \bar{x}_n \hat{\beta}_1) - \frac{1}{n} \sum_{i=1}^n x_i y_i = 0 \quad (3.32)$$

which gives us

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - n \bar{x}_n \bar{y}_n}{\sum_{i=1}^n (x_i - \bar{x}_n)^2}. \quad (3.33)$$

Note that (3.33) is just a reformation of the result, since one have

$$\sum_{i=1}^n (x_i - \bar{x}_n)(y_i - \bar{y}_n) = \sum_{i=1}^n x_i y_i - \bar{y}_n \sum_{i=1}^n x_i - \bar{x}_n \sum_{i=1}^n y_i + n \bar{x}_n \bar{y}_n \quad (3.34)$$

$$= \sum_{i=1}^n x_i y_i - n \bar{y}_n \bar{x}_n - n \bar{y}_n \bar{x}_n + n \bar{x}_n \bar{y}_n \quad (3.35)$$

$$= \sum_{i=1}^n x_i y_i - n \bar{x}_n \bar{y}_n \quad (3.36)$$

and

$$\sum_{i=1}^n (x_i - \bar{x}_n)^2 = \sum_{i=1}^n x_i^2 + n \bar{x}_n^2 - 2 \bar{x}_n \sum_{i=1}^n x_i \quad (3.37)$$

$$= \sum_{i=1}^n x_i^2 + n \bar{x}_n^2 - 2 \bar{x}_n^2 \quad (3.38)$$

$$= \sum_{i=1}^n x_i^2 - n \bar{x}_n^2. \quad (3.39)$$

Since such $(\hat{\beta}_0, \hat{\beta}_1)$ is the unique point that cancels the gradient, followed by the fact that $\mathcal{O}_{LS}$ is strictly convex, so such $(\hat{\beta}_0, \hat{\beta}_1)$ is the unique minimizer, where

$$\hat{\beta}_0 = \hat{\beta}_0 = \bar{y}_n - \bar{x}_n \hat{\beta}_1 \quad \text{and} \quad \hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}_n)(y_i - \bar{y}_n)}{\sum_{i=1}^n (x_i - \bar{x}_n)^2}. \quad (3.40)$$

■

Conceptually, the proof is not hard but it requires a lot of computations. One can also use the matrix representation form but that would be more complicated since it involves inverting a matrix.

Proposition

Proposition 11. *Let the design matrix X be full rank and assume X is orthogonal, then the coefficients of $\hat{\beta}_n$ are equal to those obtained by fitting p independent simple linear regressions.*

We need a remark from linear algebra for this proposition:

REMARK

Suppose $\mathbf{X}$ is an orthonormal matrix, then $\mathbf{X}^\top \mathbf{X} = \mathbb{I}_p$. If normal condition is dropped and we only have orthogonal matrix, then

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} \sum_{i=1}^p X_{i1}^2 & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \sum_{i=1}^p X_{ip}^2 \end{pmatrix} \quad (3.41)$$

Corollary

Corollary 1. (Indicator Regression) Let $\mathcal{D} := \{i \in [n] : (\mathbf{x}_i, y_i)\} \subseteq \mathbb{R}^p \times \mathbb{R}$ be the set of observations, let $T \in [n-1]$ be fixed, $\{A_j\}_{j \in [T]}$ be a partition of $\mathbb{R}^p$, define the map $\mathcal{M} : \mathbb{R}^p \rightarrow \mathbb{R}^T$ by

$$\mathbf{x}_i \mapsto \mathbf{z}_i := [\mathbb{1}_{A_1}(\mathbf{x}_i) \ \cdots \ \mathbb{1}_{A_T}(\mathbf{x}_i)]^\top \quad (3.42)$$

then the OLS estimator $\hat{\beta}_z := (\hat{\beta}_{z1}, \dots, \hat{\beta}_{zT})^\top \in \mathbb{R}^T$ of the regression function of y on $\mathbf{z}$ is

$$\hat{\beta}_{zj} := \frac{\sum_{i=1}^n \mathbb{1}_{A_j}(\mathbf{x}_i) y_i}{\sum_{i=1}^n \mathbb{1}_{A_j}(\mathbf{x}_i)}. \quad (3.43)$$

Proof. We know the normal equation is defined by $\mathbf{Z}^\top \mathbf{Z} \hat{\beta}_T = \mathbf{Z}^\top \mathbf{y}$ where the design matrix $\mathbf{Z}$ takes the form

$$\mathbf{Z} = \begin{pmatrix} \mathbb{1}_{A_1}(\mathbf{x}_1) & \cdots & \mathbb{1}_{A_T}(\mathbf{x}_1) \\ \vdots & \ddots & \vdots \\ \mathbb{1}_{A_1}(\mathbf{x}_n) & \cdots & \mathbb{1}_{A_T}(\mathbf{x}_n) \end{pmatrix} \in M_{n \times T}(\mathbb{R}) \quad (3.44)$$

Hence the normal equation can be written as

$$\begin{pmatrix} \mathbb{1}_{A_1}(\mathbf{x}_1) & \cdots & \mathbb{1}_{A_1}(\mathbf{x}_n) \\ \vdots & \ddots & \vdots \\ \mathbb{1}_{A_T}(\mathbf{x}_1) & \cdots & \mathbb{1}_{A_T}(\mathbf{x}_n) \end{pmatrix} \cdot \begin{pmatrix} \mathbb{1}_{A_1}(\mathbf{x}_1) & \cdots & \mathbb{1}_{A_T}(\mathbf{x}_1) \\ \vdots & \ddots & \vdots \\ \mathbb{1}_{A_1}(\mathbf{x}_n) & \cdots & \mathbb{1}_{A_T}(\mathbf{x}_n) \end{pmatrix} \cdot \begin{pmatrix} \hat{\beta}_{z1} \\ \vdots \\ \hat{\beta}_{zT} \end{pmatrix} \quad (3.45)$$

$$= \begin{pmatrix} \mathbb{1}_{A_1}(\mathbf{x}_1) & \cdots & \mathbb{1}_{A_1}(\mathbf{x}_n) \\ \vdots & \ddots & \vdots \\ \mathbb{1}_{A_T}(\mathbf{x}_1) & \cdots & \mathbb{1}_{A_T}(\mathbf{x}_n) \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \quad (3.46)$$

which simplifies to

$$\text{diag} \left(\sum_{i=1}^n \mathbb{1}_{A_1}(\mathbf{x}_i), \dots, \sum_{i=1}^n \mathbb{1}_{A_T}(\mathbf{x}_i) \right) \begin{pmatrix} \hat{\beta}_{z1} \\ \vdots \\ \hat{\beta}_{zT} \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n \mathbb{1}_{A_1}(\mathbf{x}_i) y_i \\ \vdots \\ \sum_{i=1}^n \mathbb{1}_{A_T}(\mathbf{x}_i) y_i \end{pmatrix} \quad (3.47)$$

Where (if) the diagonal matrix is invertible, then the coefficient $\hat{\beta}_{zj}$ is given by

$$\hat{\beta}_{zj} := \frac{\sum_{i=1}^n \mathbb{1}_{A_j}(x_i) y_i}{\sum_{i=1}^n \mathbb{1}_{A_j}(x_i)}. \quad (3.48)$$

■

3.2 Hat Matrix and Annihilator Matrix

Note that although we have obtained the estimate $\hat{\beta}$, but our final goal is to say something about β . However β is never observed! So our goal would be to find an estimator $\hat{\beta}$ with low variance and low bias. The figure below illustrates two different estimates: One with low variance but high bias, one with high variance but low bias:

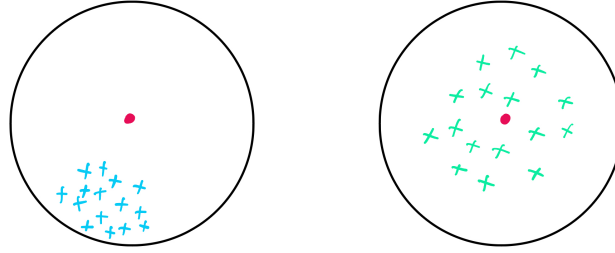


Figure 2: In the figure above, the red dot is our target parameter and crosses marked in blue and green are two estimates. The first one (left) has low variance but high bias; the second one (right) has high variance but low bias.

Once we have obtained the OLS estimator $\hat{\beta}_n$, the predicted value at a new given point $x \in \mathbb{R}^p$ is now given by

$$\hat{m}(x) := x^\top \hat{\beta}. \quad (3.49)$$

Definition

Definition 10. When $x = x_i$ for some $i \in [n]$, we say $\hat{m}(x_i) = \hat{y}_i$ is the fitted value of element i , and the OLS sample residual of element $i \in [n]$ is defined as $\hat{e}_i := y_i - \hat{y}_i$. Additionally we define the hat matrix to be $H = X(X^\top X)^{-1}X^\top$ and the annihilator matrix to be $M := I - H$.

Then, we see that the fitted values $\hat{y} := (\hat{y}_1, \dots, \hat{y}_n)^\top$ can be written as $\hat{y} = Hy$. That's probably why we call it a "Hat" matrix since it puts a hat on y

Proposition

Proposition 12. The hat matrix H and the annihilator matrix M are both orthogonal projection matrices (idempotent and symmetric).

Remark: Idempotent means we have $A^2 = A$, as for symmetric, you know that.

Proof. We prove it for $\mathbf{H}$ first. Note that

$$\mathbf{H}^\top = \left(\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \right)^\top = (\mathbf{X}^\top)^\top \left((\mathbf{X}^\top \mathbf{X})^{-1} \right)^\top \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \quad (3.50)$$

where we used the fact that $(ABC)^\top = C^\top B^\top A^\top$, and $(A^{-1})^\top = (A^\top)^{-1}$ (given A is invertible). Then,

$$\mathbf{H}^2 = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \underbrace{\mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}}_{\text{note that this is } \mathbb{I}} \mathbf{X}^\top := \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top. \quad (3.51)$$

Hence $\mathbf{H}$ is an orthogonal projection, and it follows that $\mathbf{M} = \mathbb{I} - \mathbf{H}$ is also an orthogonal projection. ■

Proposition

Proposition 13. $\mathbf{H}$ projects onto $\text{Im}(\mathbf{X})$ and $\mathbf{M}$ projects onto $\text{Ker}(\mathbf{X}^\top)$;

Proof. We see that $\forall \mathbf{y} \in \mathbb{R}^n$, denote $\hat{\boldsymbol{\beta}}_n$ to be the OLS estimate, then by definition we have indeed

$$\mathbf{H}\mathbf{y} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \in \text{Im}(\mathbf{X}) \quad (3.52)$$

We further use the fact that $\text{Im}(\mathbf{X}^\perp) = \text{Ker}(\mathbf{X}^\top)$ to conclude $\mathbf{M} := \mathbb{I} - \mathbf{H}$ is the projection onto $\text{Ker}(\mathbf{X}^\top)$. Since if $\mathbf{X}$ projects onto S then $\mathbb{I} - \mathbf{X}$ projects onto $S^\perp$. ■

Proposition

Proposition 14. $\text{tr}(\mathbf{H}) = p, \text{tr}(\mathbf{M}) = n - p$. (Recall that p is the number of covariates and n is the number of sample)

Proof. We use the fact that $\text{tr}(AB) = \text{tr}(BA)$. So we have

$$\text{tr}(\mathbf{H}) = \text{tr}(\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top) = \text{tr}(\mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}) = \text{tr}(\mathbb{I}_p) = p. \quad (3.53)$$

Note that $\mathbf{X} \in M_{n \times p}(\mathbb{R}), \mathbf{H} \in M_{n \times n}(\mathbb{R})$. Also we have $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$, so we have

$$\text{tr}(\mathbf{M}) = \text{tr}(\mathbb{I}_n - \mathbf{H}) = \text{tr}(\mathbb{I}_n) - \text{tr}(\mathbf{H}) = n - p. \quad (3.54)$$

■

Proposition

Proposition 15. Suppose intercept is included in the covariates, then $(\hat{\mathbf{y}} - \bar{\mathbf{y}})^\top \hat{\mathbf{e}} = 0$, where

$$\hat{\mathbf{y}} = (\hat{y}_1, \dots, \hat{y}_n)^\top, \bar{\mathbf{y}} = (\bar{y}_1, \dots, \bar{y}_n)^\top, \hat{\mathbf{e}} = (\hat{e}_1, \dots, \hat{e}_n)^\top. \quad (3.55)$$

Proof. We already know that $\hat{\mathbf{y}} \in \text{Im}(\mathbf{H})$, $\hat{\mathbf{e}} \in \text{Im}(\mathbf{M})$ and it is clear that $\text{Im}(\mathbf{H})$ and $\text{Im}(\mathbf{M})$ are orthogonal to each other hence $\hat{\mathbf{y}}^\top \hat{\mathbf{e}} = 0$ and we only need to show $\hat{\mathbf{y}} \in \text{Im}(\mathbf{H})$. Since intercept is included, denote the design matrix $\mathbf{X}$ by

$$\mathbf{X} = \begin{pmatrix} 1 & x_{12} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n2} & \cdots & x_{np} \end{pmatrix} \quad (3.56)$$

hence $(1, \dots, 1)^\top \in \text{Im}(\mathbf{X})$, thus $(1, \dots, 1)^\top \in \text{Im}(\mathbf{H})$ (recall that $\mathbf{H}$ projects onto $\mathbf{X}$), and so $\hat{\mathbf{y}}_n \in \text{Im}(\mathbf{H})$ since it's just a difference by a scalar. ■

Definition

Definition 11. The leverage of element $i \in [n]$ is defined

$$h_{ii} := \mathbf{H}_{(ii)} := \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i \quad (3.57)$$

and an extension yields

$$h_{ij} := \mathbf{H}_{(ij)} = \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_j \quad (3.58)$$

Leverages appear in many places in linear regression analysis, especially when p is large. We study some properties of leverage:

Corollary

Corollary 2. Leverage satisfies the following properties:

- (i) The leverage are positive and bounded ($0 \leq h_{ii} \leq 1$);
- (ii) The leverages are symmetric, i.e $h_{ij} = h_{ji}$;
- (iii) For any $j \in [n]$, $\hat{\mathbf{y}}_j = \sum_{i \in [n]} h_{ij} y_i$;
- (iv) If intercept is included, then denote $\mathbf{x} = (1, x_{(1)}, \dots)^\top$, then

$$\sum_{i \in [n]} \mathbf{x}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i = 1 \quad (3.59)$$

and in particular, $\sum_{i \in [n]} h_{ij} = 1$.

We will also study

3.3 Properties of the OLS estimator

We will investigate several properties of the OLS estimator given by

$$\hat{\beta}_n := (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \quad (3.60)$$

Theorem

Theorem 13. *The OLS estimator is unbiased given $\mathbf{X}$, that is,*

$$\mathbb{E}[\hat{\beta}_n | \mathbf{X}] = \beta. \quad (3.61)$$

Proof. Note that we have

$$\mathbb{E}[\hat{\beta}_n | \mathbf{X}] = \mathbb{E}[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} | \mathbf{X}] \quad (3.62)$$

$$= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}[\mathbf{y} | \mathbf{X}] \quad (3.63)$$

$$= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}[\mathbf{X}\beta + \mathbf{e} | \mathbf{X}] \quad (3.64)$$

$$= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \mathbb{E}[\beta | \mathbf{X}] \quad (3.65)$$

$$= \beta. \quad (3.66)$$

where we used the fact that $\mathbb{E}[\mathbf{e} | \mathbf{X}] = \mathbf{0}$. ■

Theorem

Theorem 14. *The covariance of the OLS estimator is given by*

$$\mathbb{V}(\hat{\beta}_n | \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \quad (3.67)$$

where $\sigma^2 := \mathbb{V}(\varepsilon) := \mathbb{V}(y | x)$.